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Experimental and numerical investigation of aeroacoustic installation effects on small-scale multi-rotor systems in hover

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https://polemecanique.github.io/internal_seminars/

Abstract

This study investigates aeroacoustic installation effects in small-scale multi-rotor systems in hover through a combined experimental and numerical approach. Measurements were conducted in an anechoic chamber for three configurations: an isolated propeller, a propeller with its supporting arm, and a tandem propeller arrangement representative of a quadcopter layout. The propellers operate at very low Reynolds numbers, around 3×10^4 , where transitional flow and viscous effects are important. A mid-fidelity aerodynamic model based on a NonLinear Vortex Lattice Method is considered to predict the aerodynamic forces used as input to Hanson's tonal noise model. The aerodynamic simulation is shown to be very sensitive to airfoil polar data. In the tandem configuration, unsteady loading noise plays a significant role. The results show that a small variation of rotational speed of rotors induces a change of directivity for the unsteady loading noise and the total noise.